Roll No.													

AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

Department of Computer Science and Engineering

Pre-University Test

Course: B. Tech. **Semester: IV**
Session: 2025-26 **Section: All 2nd year sections (CSE & IT and allied branches)**

Subject: Object Oriented Programming with Java **Sub. Code: BCS403**
Max Marks: 70 **Time: 3 hrs.**

OBE Remarks:

Q.No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
CO No.	1	1	2	2	3	4	5	1	2	2	2	3	3	4	4	5	5
Bloom's Level	L1	L2	L1	L2	L1	L2	L1	L3	L3	L6	L5	L6	L3	L2	L5	L4	L3
Weightage CO3: 16							Weightage CO4: 16							Weightage CO5: 16			

Note: Answer **all** the sections.

Section-A

A. Attempt **all** the parts. **(7 X 2 =14)**

1. What is the purpose of static variable?
2. Describe interface of Java.
3. Differentiate between throw and throws clauses.
4. How to create the object of BufferedReader class.
5. Define text block with an example.
6. Difference between Comparator and Comparable interfaces.
7. What do mean by Autowiring?

Section-B

B. Attempt **Any three. (Q. No. 12 is Compulsory)** **(3X 7=21)**

8. Describe the creating and using the package in Java program with suitable examples.
9. Create a program to demonstrate overloading and overriding features of Java. Explain the program.
10. Create a Java program for handling exceptions. Explain the program. Write the differences between checked and unchecked exceptions.
11. Discuss any one method of implementing multithreading with an example and evaluate importance of using that method.
12. Create a program in Java to generate and display a random number using functional interface and lambda expression. Also describe functional interface and lambda expression concept with examples.

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Section-C

C. Attempt **all** the parts. **(5 X 7 = 35)**

13. Attempt any one.

a) Explain Base64 Encode and Decode mechanism. Apply Base64 conversion in a Java program by taking a string message as an example.

b) Discuss the concepts of Switch Expression and Local Variable Type Inference with an example of each.

14. Attempt any one.

c) Discuss List, Set and SortedSet Collection interfaces with their important methods.

d) What is Enumeration interface? Discuss legacy class Vector with an example.

15. Attempt any one.

a) Create a TreeMap class program in Java and evaluate the importance of using TreeMap. Also write syntax for Map interface.

b) Demonstrate ArrayList Collection class with an example. Evaluate the utility of ArrayList class.

16. Attempt any one.

c) Describe different ways of Dependency Injection (DI) in the Spring Framework with example of each. Analyze benefits of DI.

d) Discuss Request and Application bean scopes of Spring Framework with examples. Analyze the difference between both the scopes.

17. Attempt any one.

e) Create brief code snippets for showing CRUD operations of RESTful Web services in Spring Boot.

f) Explain bean life cycle in Spring Boot with a suitable diagram and code snippets.

Faculty Sign

HoD Sign